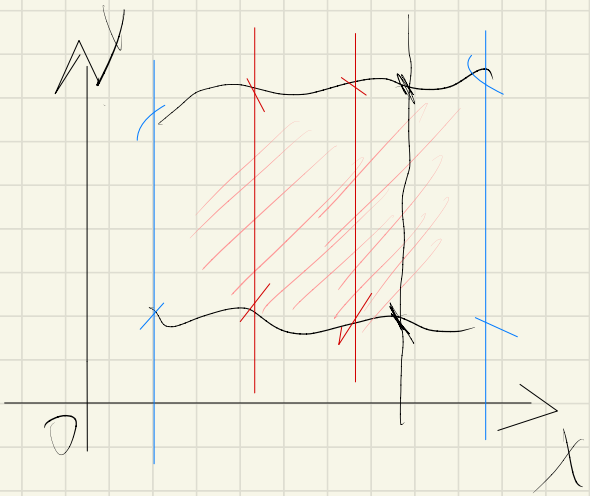
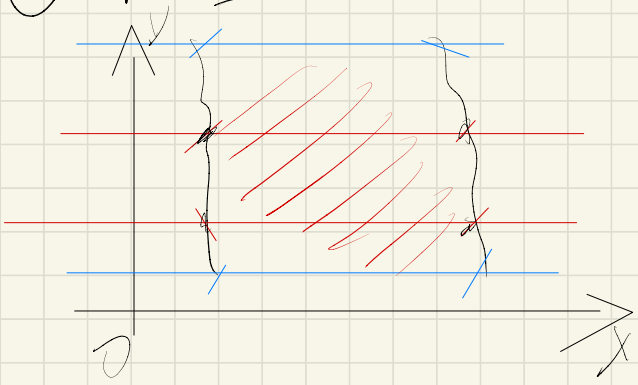
积分区域的分类：

① X-型区域：



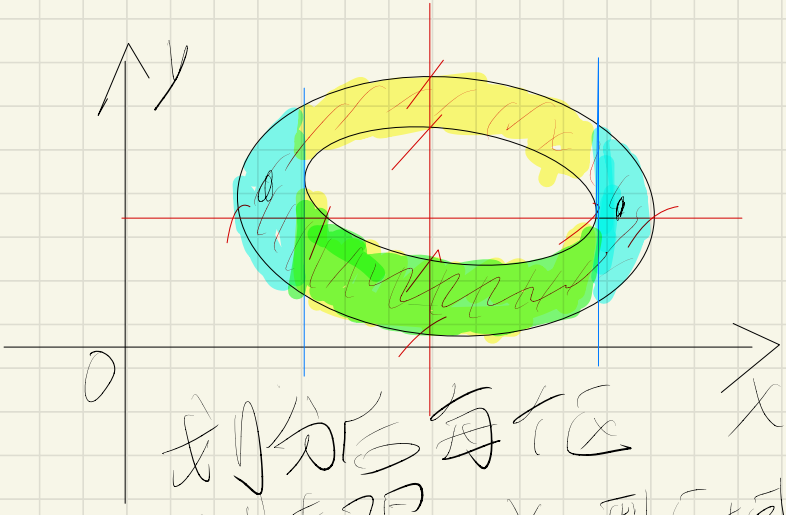
从任意位置作平行于y轴的直线只有两个交点

② Y-型区域：



在任意位置作平行于x轴的直线与区域的交线只有两个

③ 非X 非Y型区域(划分)



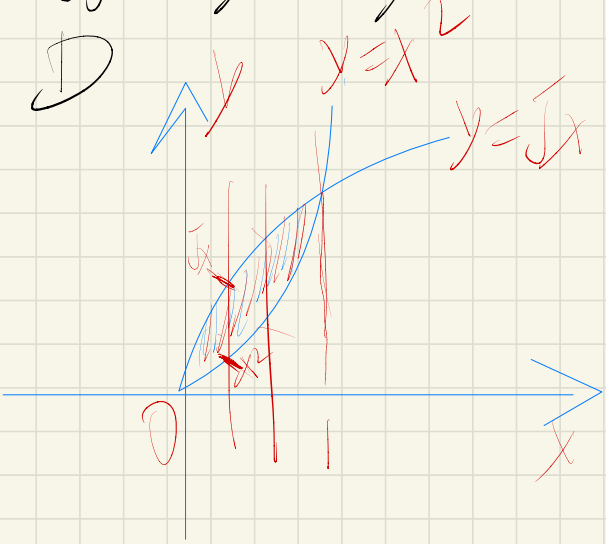
显然两个方向的交点数量均2

划分后每个区域都是 X-型区域

二重积分的计算

eg.1：设D由$y=x^2$和$y=\sqrt{x}$所围，求

$$\iint_D x^2 y \, dx dy$$



$x^2=\sqrt{x} \Rightarrow x=1$

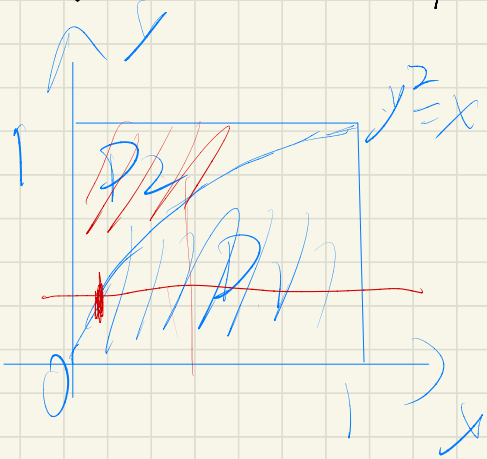
$$\int_0^1 dx \int_{x^2}^{\sqrt{x}} x^2 y \, dy$$

$$= \int_0^1 \frac{x^2}{2} y^2 \Big|_{x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 \frac{x^3}{2} - \frac{x^6}{2} dx$$

$$= \frac{3}{64\times 7}$$

例2：设 $D:\begin{cases}0\le x\le 1\\ 0\le y\le 1\end{cases}$ 求 $\iint_D |y^2-x|\,dxdy$



$$\begin{cases}y^2\ge x\\ y^2\le x\end{cases}$$

原式 $=\iint_{D_1} x-y^2\,dxdy+\iint_{D_2} y^2-x\,dxdy$

$=\int_0^1 dy\int_{y^2}^1 x-y^2\,dx$

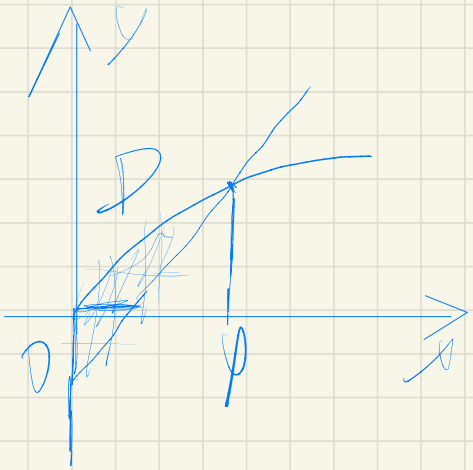
$+\int_0^1 dy\int_0^{y^2} y^2-x\,dx$

$=\int_0^1\left(\frac{1}{2}x^2-y^2x\Big|_{y^2}^1\right)dy=\int_0^1 \frac{1}{2}-y^2+\frac{1}{2}y^4\,dy$

$+\int_0^1\left(y^2x-\frac{1}{2}x^2\Big|_0^{y^2}\right)dy=\int_0^1 \frac{1}{2}y^4\,dy$

例3：设D由$y=x$和$y=x-2$所围成，求

$$I=\iint_D xy\,dxdy$$



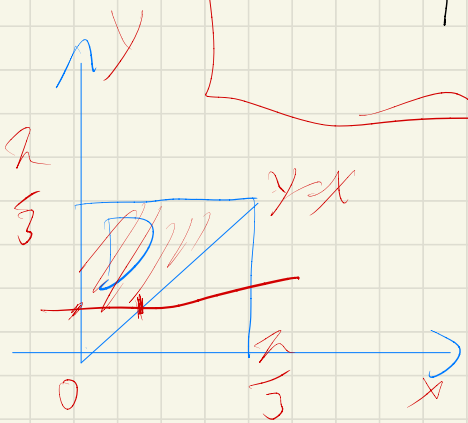
$y=x-2$

$x=\varphi$

$$\int_0^{\varphi}dx\int_{x-2}^{\sqrt{x}}xy\,dy$$

$$=\int_0^{\varphi}\left(\frac{y^2}{2}x\Big|_{x-2}^{\sqrt{x}}\right)dx$$

例4：$D:\begin{cases}0\le x\le \frac{\pi}{3}\\ x\le y\le \frac{\pi}{3}\end{cases}$，求 $\iint_D \frac{\sin y}{y}\,dxdy$



$$\int_0^{\frac{\pi}{3}} dx \int_x^{\frac{\pi}{3}} \frac{\sin y}{y}\,dy$$

原函数不存在

$$\int_0^{\frac{\pi}{3}} dy \int_0^{y} \frac{\sin y}{y}\,dx$$

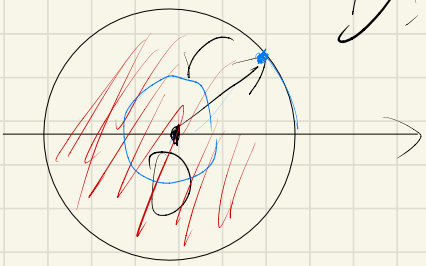
$$\int_0^{\frac{\pi}{3}} \frac{\sin y}{y}\cdot x\Big|_0^{y}\,dy$$

$$=\int_0^{\frac{\pi}{3}} \sin y\,dy = -\cos y\Big|_0^{\frac{\pi}{3}}$$

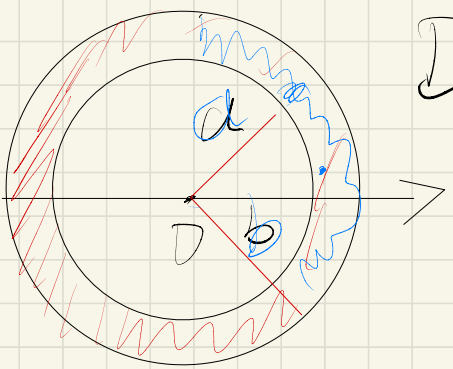
$$=\frac{1}{2}$$

二重积分的极坐标和广义极坐标形式

① 极坐标的圆和圆环、椭圆、直线



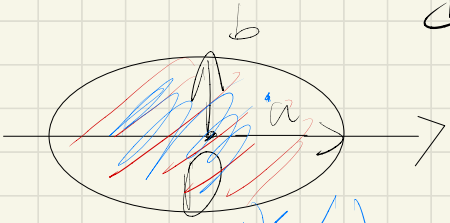
$$D:\begin{cases}0\le\rho\le r\\ 0\le\theta\le 2\pi\end{cases}\Rightarrow\begin{cases}\rho\sin\theta=y\\ \rho\cos\theta=x\end{cases}$$



$$D:\begin{cases}a\le\rho\le b\\ 0\le\theta\le 2\pi\end{cases}$$

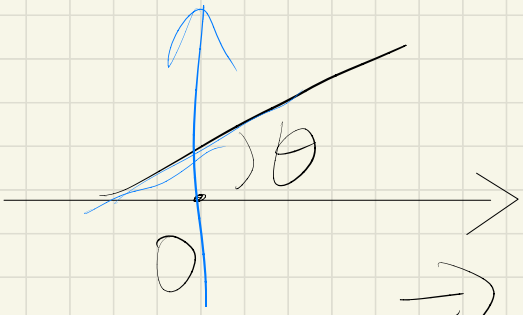
$\rho^2\le 1$

$0\le\rho\le 1$



$$D:\begin{cases}0\le\rho\le 1\\ 0\le\theta\le 2\pi\end{cases}\Rightarrow\begin{cases}b\rho\sin\theta=y\\ a\rho\cos\theta=x\end{cases}$$

$\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1$



若 $l: kx+b=y$

$\Rightarrow \rho k\sin\theta + b = \rho\cos\theta$

$\Rightarrow \begin{cases} x=\rho\cos\theta \\ y=\rho\sin\theta \end{cases}$

公式：

$\rho(k\sin\theta-\cos\theta)=-b \Rightarrow$

$$\iint_D f(x,y)\,dx\,dy = \iint_{D'} f(\rho,\theta)\cdot\rho\,d\rho\,d\theta$$

$f(\rho\cos\theta, \rho\sin\theta)$

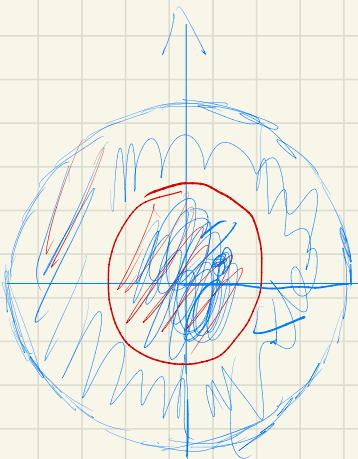
e.g. $\iint_D (1-x)\,|x^2+y^2-\varphi|\,dx\,dy$,

$D=\{(x,y)\mid (x^2+y^2)\le 16\}$

$\begin{cases} x^2+y^2\ge\varphi \\ x^2+y^2\le 4 \end{cases}$ →

$= \iint_D |x^2+y^2-\varphi|\,dx\,dy$

$\begin{cases} x^2+y^2\ge\varphi \\ x^2+y^2\le\varphi \end{cases}$



$D_1 \cup D_2 = D$

$\begin{cases} 0\le\rho\le 2 \\ 0\le\theta\le 2\pi \end{cases}$

$\iint_{D_1} 4-(x^2+y^2)\,dx\,dy$

$\int_0^{2\pi}\int_0^2 \left(4-(\rho^2\cos^2\theta+\rho^2\sin^2\theta)\right)\rho\,d\rho\,d\theta$

$\begin{cases} x=\rho\cos\theta \\ y=\rho\sin\theta \end{cases}$

$$\iint_D \rho(4-\rho^2)\,d\rho\,d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^2 \rho(4-\rho^2)\,d\rho$$

$$= 2\pi \int_0^2 \rho(4-\rho^2)\,d\rho$$

$$= 2\pi \left(2\rho^2 - \frac{1}{4}\rho^4\right)\Big|_0^2$$

$$= 2\pi(8-4) = 8\pi$$

$$\iint_{D_2} (\rho^2\cos^2\theta + \rho\sin\theta - \varphi)\,\rho\,d\rho\,d\theta$$

$$= \int_0^{2\pi} d\theta \int_2^4 (\rho^2-4)\rho\,d\rho$$

$$= 2\pi\left(\frac{1}{4}\rho^4 - 2\rho^2\right)\Big|_2^4 = 2\pi\left[(64-32)-(4-8)\right]$$

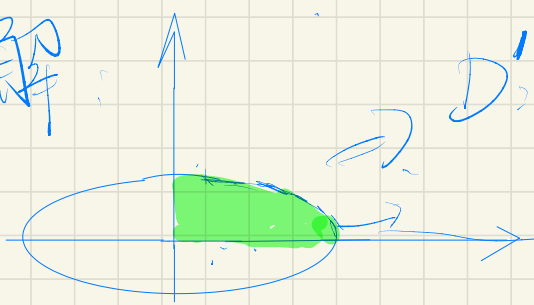
$$= 72\pi$$

公式：

$$\iint_D f(x,y)\,dxdy = \iint_{D'} ab\rho f(\rho,\theta)\,d\rho d\theta$$

e.g.2 $\iint_D (x^2+y^2)\,dxdy$, $D=\begin{cases}\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1\\ x\ge 0, y\ge 0\end{cases}$

解：



$\Rightarrow D'\begin{cases}0\le\theta\le\frac{\pi}{2}\\ \rho^2\le 1\end{cases}$ $0\le\rho\le 1$

ρ 与 θ 无关

$$\iint_D x^2+y^2\,dxdy = \iint_{D'} ab\rho(a^2\rho^2\cos^2\theta + b^2\rho^2\sin^2\theta)\,d\rho d\theta$$

$$= \int_0^{\frac{\pi}{2}}(a^2\cos^2\theta + b^2\sin^2\theta)\,d\theta \int_0^1 ab\rho^3\,d\rho$$

$$a^2 \int_0^{\frac{\pi}{2}} \cos^2\theta \, d\theta + b^2 \int_0^{\frac{\pi}{2}} \sin^2\theta \, d\theta$$

$$= a^2 \cdot \frac{1}{2} \times \frac{\pi}{2} + b^2 \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= \frac{\pi}{4}(a^2 + b^2) \cdot \frac{1}{4} \rho^4 \, ab \, |b|$$

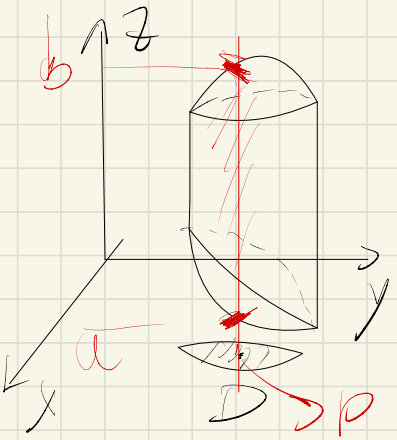
$$= \frac{\pi (a^2+b^2) ab}{n}$$

先二后一的积分

$$\int_a^b dz \iint_D$$

三重积分的计算

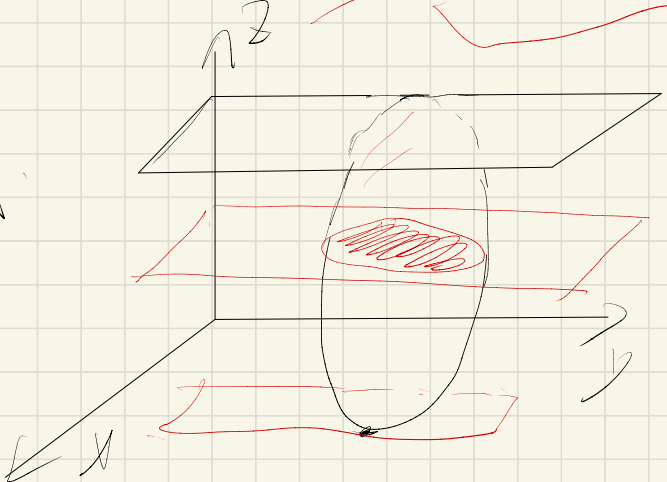
①柱型域



在投影D中任取一点P，作直线平行z轴，交线为一段线段，称为Z型柱

X型柱，Y型柱

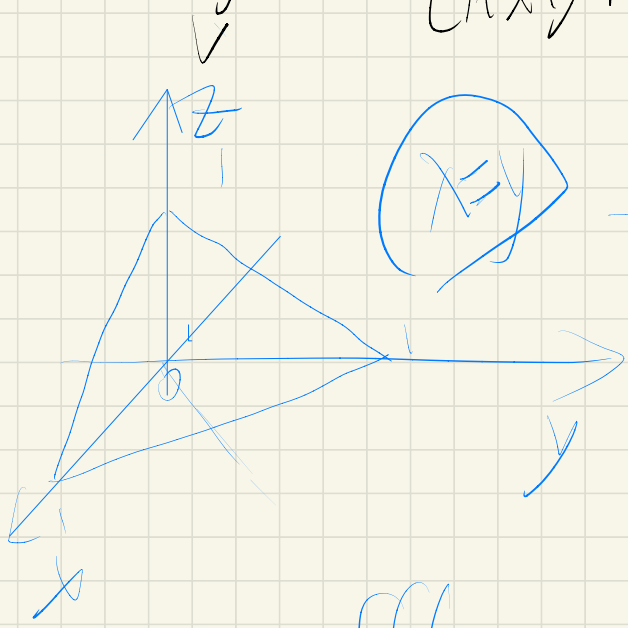
②片型域



z轴同高的第一片区域

e.g.1：设V是平面$x+y+z=1$与三个坐标轴平面所围成的四面体，求

$$I=\iiint_V \frac{1+\sin^2x\sin^2y}{(1+x+y+z)^2}\,dxdydz$$



$$\begin{cases}0\le x\le 1\\ 0\le y\le 1-x\\ 0\le z\le 1-x-y\end{cases}$$

$$\iiint_V x\,dxdydz=\iiint_V y\,dxdydz$$

$$\int_a^b dz \iint_{D(z)} \frac{\sin^2 x}{(1+x+y+z)^2}\,dx\,dy\,dz$$

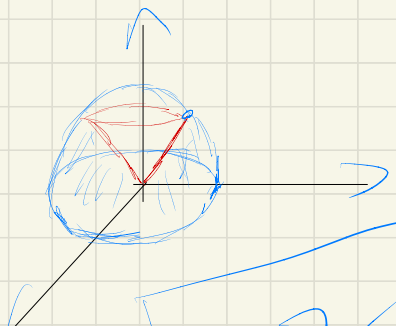
$$\int_a^b dz \iint_{D(z)} \frac{\sin^2 y}{(1+x+y+z)^2}\,dx\,dy\,dz$$

$$\int_a^b dz \iint_{D(z)} \frac{1}{(1+x+y+z)^2}\,dx\,dy\,dz$$

$$\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} \frac{1}{(1+x+y+z)^2}\,dz$$

e.g. 2 设 Ω 由 $z=\sqrt{x^2+y^2}$ 和半球面 $z=\sqrt{1-x^2-y^2}$ 所围成的空间立体，

求 $\iiint_\Omega z\,dx\,dy\,dz$

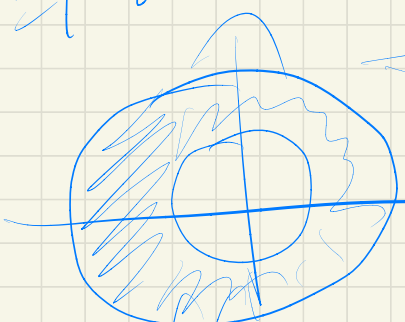


$$\sqrt{x^2+y^2} = \sqrt{1-x^2-y^2}$$

$$\Rightarrow x^2+y^2 = \frac{1}{2}$$

$$\begin{cases} 0 \le z \le \frac{\pi}{2} \\ z = \frac{\pi}{2} \end{cases}$$

$$x^2+y^2 \le 1-z^2 \qquad z \ge x^2+y^2$$



$$\int_0^{\frac{\pi}{2}} z\,dz \iint_{D_z} dx\,dy$$

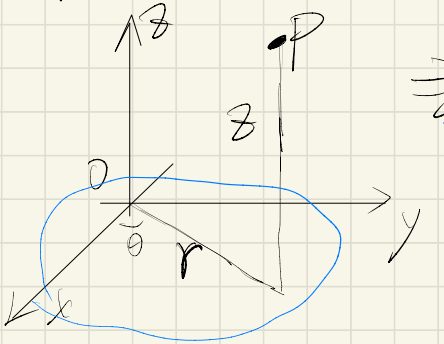
$$\pi(1-z^2) \qquad \pi(z^2)$$

$$\int_0^{\frac{\pi}{2}} \pi(1-2z^2)\,z\,dz$$

$$\frac{\pi}{2}z^2 - \frac{\pi}{2}z^4 \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{8}$$

三重积分在球坐标和柱坐标下的形式

1. 柱坐标：



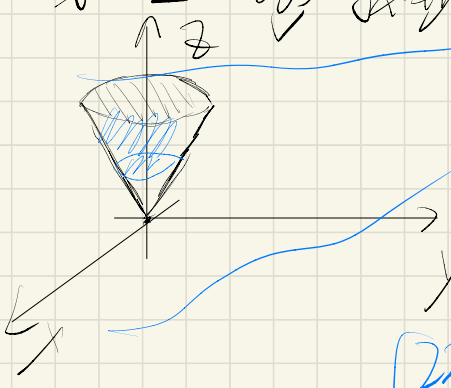
$\Rightarrow P(r,\theta,z)$

$\Rightarrow \begin{cases} x=r\cos\theta \\ y=r\sin\theta \\ z=z \end{cases}$

公式：$\iiint f(x,y,z)\,dxdydz=\iiint f(r\cos\theta,r\sin\theta,z)\,drd\theta dz$

eg.1：设V由$z=\sqrt{x^2+y^2}$，平面$z=1$和$z=2$所围

求 $I=\iiint \frac{e^{z^2}}{\sqrt{x^2+y^2}}dxdydz$



$\sqrt{x^2+y^2}=r$

$\begin{cases} 0\le\theta\le 2\pi \\ 1\le z\le 2 \\ 0\le r\le z \end{cases}$

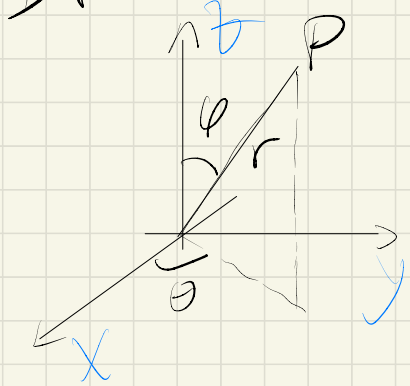
$\int_0^{2\pi}d\theta\int_1^2 dz\int_0^z r\cdot\frac{e^{z^2}}{r}dr$

$$\int_0^{2\pi} d\theta \int_1^2 z e^{z^2} dz$$

$$= \frac{2\pi}{2} \int_1^2 e^{z^2} dz^2$$

$$= \pi e^{z^2} \Big|_1^2 = \pi (e^4 - e)$$

球坐标：



$$P(r, \varphi, \theta)$$

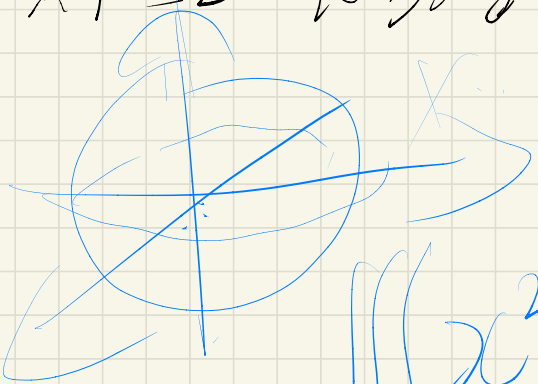
$$\Rightarrow \begin{cases} z = r\cos\varphi \\ x = r\sin\varphi\cos\theta \\ y = r\sin\varphi\sin\theta \end{cases}$$

公式：$\iiint_{\Omega} f(x,y,z)\,dxdydz$

$= \iiint_{\Omega} r^2\sin\varphi\, f(r\sin\varphi\cos\theta, r\sin\varphi\sin\theta, r\cos\varphi)\, drd\theta d\varphi$

e.g. 2 计算 $I=\iiint_{\Omega}(4x^2+5y^2+6z^2-10xy)\,dxdydz$

其中 $\Omega=\{(x,y,z)\mid x^2+y^2+z^2\le 1\}$



$\iiint_{\Omega} x^2\,d\Omega$

$= \iiint_{\Omega} y^2\,d\Omega = \iiint_{\Omega} z^2\,d\Omega$

$= \frac{1}{3}\iiint_{\Omega}(x^2+y^2+z^2)\,d\Omega$ $\{\rho^2$

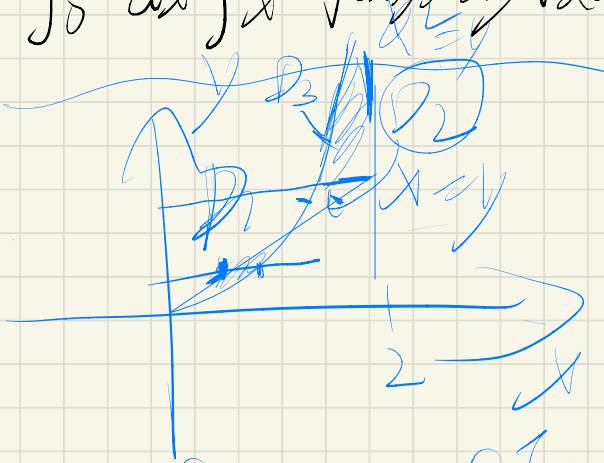
$= \int_0^{2\pi} d\theta \int_0^1 \rho^4 d\rho$

$= 2\pi \times \frac{1}{5}$

$= \frac{2\pi}{5}$

累次积分的换序：

e.g.1 (二重积分换序) 设 $f(x,y)$ 连续，请将 $\int_0^2 dx \int_x^{x^2} f(x,y)\,dy$ 换序



$x=y \quad x=\sqrt{y}$

$x=x^2 \Rightarrow x=1$

$y=1$

$\int_0^1 dy \int_y^{\sqrt{y}} f(x,y)\,dx$

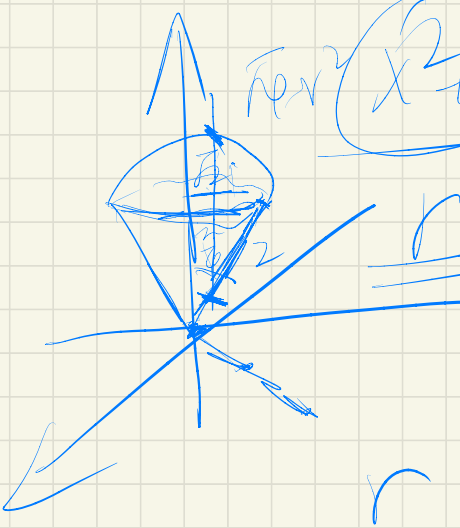
$+ \int_1^2 dy \int_y^2 f(x,y)\,dx$

$$+\int_2^{\varphi} dy \int_{\frac{y}{2}}^{2} f(x,y)\,dx$$

eg2 将球坐标系下的三次积分

$$\int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{2} f(\rho\sin\varphi\cos\theta, \rho\sin\theta\sin\varphi, \rho\cos\varphi) \cdot \rho^2\sin\varphi \, d\rho\, d\theta\, d\varphi$$

, 换柱坐标系下的积分形式



$x^2+y^2+z^2 \le \varphi$

$r^2+z^2 \le \varphi$

r

$$\begin{cases} 0 \le \rho \le 2 \\ 0 \le \varphi \le \frac{\pi}{4} \\ 0 \le \theta \le 2\pi \end{cases}$$

$$\begin{cases} 0 \le \theta \le 2\pi \\ 0 \le r \le \sqrt{2} \\ r \le z \le \sqrt{\varphi - r^2} \end{cases}$$

$$\int_0^{2\pi} d\theta \int_0^{\sqrt{2}} dr \int_r^{\sqrt{\varphi-r^2}} r \cdot f(\,)\,dz$$

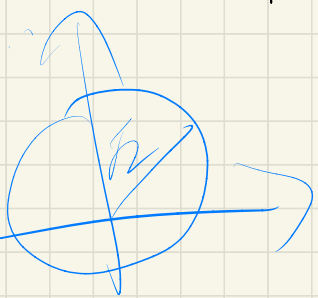
多重积分的应用：

1. 曲面面积 $S=\iint_D \sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\,dxdy$

$z=f(x,y)$

e.g.1 $z=\frac{x^2+y^2}{2}$, $0\le z\le 1$, 求壳的面积

$x^2+y^2\le 2$



$$\sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}$$

$$=\iint_D \sqrt{1+x^2+y^2}\,dxdy$$

$$=\int_0^{2\pi} d\theta \int_0^{\sqrt{2}} \sqrt{1+r^2}\,dr$$

$= \pi \int_0^{\sqrt{2}} \int r^2 \, d_3(1+r^2)$

$= \pi \cdot \frac{2}{3} \cdot (1+r^2)^{\frac{3}{2}} \Big|_0^{\sqrt{2}}$

$= 2\sqrt{3}\pi$

质心的计算

公式：

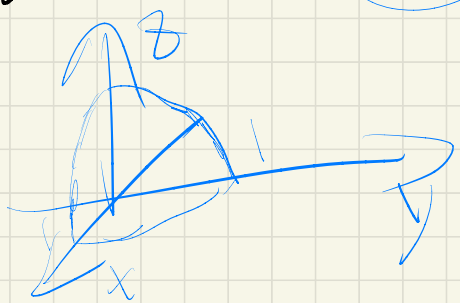
$$\bar{x} = \frac{\iiint_\Omega x \cdot u(x,y,z)\, d\Omega}{\iiint_\Omega u(x,y,z)\, d\Omega}$$

$$\bar{y} = \frac{\iiint_\Omega y \cdot u(x,y,z)\, d\Omega}{\iiint_\Omega u(x,y,z)\, d\Omega}$$

$$\bar{z} = \frac{\iiint_\Omega z \cdot u(x,y,z)\, d\Omega}{\iiint_\Omega u(x,y,z)\, d\Omega}$$

当 $u(x,y,z)$ 为常数时，又称为形心

e.g.2 求 1/8 球面的形心



① $\iiint_\Omega x \, d\Omega$

$= \iiint_\Omega y \, d\Omega = \iiint_\Omega z \, d\Omega$

$$\iint_D x \cdot \sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\,dx\,dy$$

$$z=\sqrt{1-x^2-y^2}$$

$$\begin{cases} x^2+y^2\le 1 \\ x\ge 0,\ y\ge 0 \end{cases}$$

$$\frac{\partial z}{\partial x}=\frac{-x}{\sqrt{1-x^2-y^2}}$$

$$\frac{\partial z}{\partial y}=\frac{-y}{\sqrt{1-x^2-y^2}}$$

$$\frac{x^2+y^2}{1-x^2-y^2}+1$$

$$\iint_D \frac{1}{\sqrt{1-(x^2+y^2)}}\,dx\,dy$$

$$\begin{cases} x^2+y^2\le 1 \\ x\ge 0,\ y\ge 0 \end{cases}$$

$$\Downarrow$$

$$\begin{cases} 0\le \rho\le 1 \\ 0\le \theta\le \frac{\pi}{2} \end{cases}$$

$$\int_0^{\frac{\pi}{2}} d\theta \int_0^1 \rho \frac{\rho\cos\theta}{\sqrt{1-\rho^2}} d\rho$$

3 转动惯量

公式：
$$\begin{cases} I_x = \int_\Omega (y^2+z^2)\mu(x,y,z)\,d\Omega \\ I_y = \int_\Omega (x^2+z^2)\mu(x,y,z)\,d\Omega \\ I_z = \int_\Omega (x^2+y^2)\mu(x,y,z)\,d\Omega \end{cases}$$

eg3 已知 $z=\sqrt{x^2+y^2}$ $(0\le z\le 1)$ 的面密度为 z，则其关于 z 轴的转动惯量 $I_z=?$

$$\frac{\partial z}{\partial x}\ \frac{\partial z}{\partial y}\ 1$$

$$I_z = \iint_\Sigma (x^2+y^2)\,dS$$

$$= \iint_\Sigma (x^2+y^2)^{\frac{3}{2}}\,dS$$

$$D:\begin{cases} x^2+y^2 \le 1 \\ z \ge 0 \end{cases} = \iint_D (x^2+y^2)^{\frac{3}{2}} \cdot \sqrt{2}\,dx\,dy$$

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} \qquad \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2+y^2}}$$

$$\begin{cases} 0 \le \rho \le 1 \\ 0 \le \theta \le 2\pi \end{cases} \qquad \int_0^{2\pi} d\theta \int_0^1 \rho \cdot (\rho^2)^{\frac{3}{2}}\,d\rho$$

$$= 2\pi \cdot \frac{1}{5}\rho^5 \Big|_0^1$$

$$= \frac{2\pi}{5}$$